

Phase 3 Making-of Abstract

ClimateReady Africa: Development and Critical Evaluation of a Readiness Assessment Application

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Abstract

ClimateReady Africa is a web application for the early preparation of fictional mitigation projects. It helps project developers and consultants identify available information, incomplete evidence and next actions before formal review. Users record a project profile and self-reported evidence against pathway-sensitive criteria. The system produces an explainable readiness score, completeness status, critical gaps, prioritised actions and a downloadable text report. Developed through solo, iterative Agile work, the prepared release passed local tests and static checks. This demonstrates a screening workflow, not project eligibility, host-country authorisation or verified cloud operation. Synthetic examples and uncalibrated criteria limit the findings to an academic software-engineering demonstration.

Keywords: Article 6, carbon markets, project readiness, software engineering, climate mitigation

1. Problem and solution concept

Targeting developers and consultants, the tool records an initial preparation profile for a fictional mitigation activity. Users select a carbon-market pathway and sector and enter supporting evidence. The rubric comprises five pillars and up to fourteen criteria; applicability varies by pathway, sector and intended use. Ratings distinguish missing, partial and documented preparation. A critical gate prevents unresolved essential items from being concealed by an aggregate score. Four fictional African projects illustrate different sectors and pathway choices. Results organise preparation and discussion. Evidence is self-reported; no output determines eligibility or replaces independent technical review or a host-Party decision.

2. Development method and technical design

The work began approximately one month before Phase 1 was submitted on 3 September 2026. Phase 2 followed on 17 September after specification refinement and implementation work; finalisation continued to the 26 September deadline. A solo iterative Agile approach linked requirements, small implementation increments, checks and documentation. Assessor feedback prompted a complete work-package catalogue, measurable quality targets, explicit diagram notation and one maintained specification. Earlier work-package dates and effort values are retrospective estimates, not time records. The application uses React and TypeScript with Vite, a Cloudflare Worker and D1, and Zod for data validation. Pure domain functions determine criterion applicability, calculate scores and statuses, generate actions and prepare report text. This separation keeps the assessment rules inspectable and testable.

3. Implementation and verification

Review of version 1.0.0 identified a public write path through which a visitor could change shared example data. Prepared version 1.0.1 addresses this defect by serving fictional templates as read-only data and rejecting anonymous server-side mutations. New and edited visitor projects remain in browser storage; Web Locks and project revisions help coordinate edits across tabs. Reset affects

only the current browser workspace, and clearing that browser's data removes its locally stored work. These choices make the public demonstration safer but do not provide synchronised multi-device accounts or central user storage.

Local verification on 25 September passed fifteen unit tests, linting and TypeScript checks. Four HTTP checks through a local Worker-compatible runtime confirmed that example data is returned, anonymous write requests receive HTTP 403 responses and shared content remains unchanged. The unit tests include scoring and validation cases as well as visitor-copy isolation, persistence, stale updates, reset, storage failure and corrupt data. These results establish evidence for the tested local behaviours only. A hosted browser walkthrough, Docker execution, performance measurements, selected accessibility checks, maintenance timing and rollback were not verified at the evidence cut-off. Therefore, no claim of successful cloud deployment or measured user benefit is made here.

4. Lessons learned and conclusion

The principal process lesson was to keep requirements, design choices and observed results distinct. Early plans included account management, administration and a different backend; the final academic demonstration uses code-controlled rules, local visitor workspaces and text reports. Updating the specification alongside implementation made these scope changes visible and improved traceability. A second lesson was that a successful build or unit suite is insufficient evidence for a public workflow: the shared-data defect required explicit isolation rules and regression checks. Work-package IDs helped connect planned work to requirements, while retrospective effort estimates were kept separate from verified dates and test outcomes.

The contribution is a transparent preparation workflow that links pathway-specific evidence maturity to actionable gaps. Its present limitations include fictional data, scoring weights and thresholds that have not been calibrated, a single developer, and no independent usability or domain-validation study. Claims about faster preparation or stakeholder acceptance remain hypotheses. ClimateReady Africa should therefore be interpreted as an academic decision-support prototype for structuring early project preparation. Its next evaluation should verify the hosted user journey, container setup and release process, then assess usability and the rubric with representative users and domain reviewers before making broader effectiveness claims.